Regular polytopes for four alternating groups

Human Compression project: L071 recovery

Working statement; frozen scientific review pending

The complete retained proof of enumeration by stabilizers of initial strings supplies the common theory. In this note isomorphism preserves the order of the distinguished generators. A marking is therefore counted separately from its dual unless an automorphism reverses it. This is the convention of Leemans–Mixer, *Algorithms for classifying regular polytopes with a fixed automorphism group*, Contributions to Discrete Mathematics 7(2) (2012), 105–118, Section 6 and Table 2, physical page 9.

Theorem 1. *The numbers of regular polytopes with the indicated automorphism groups, up to isomorphism, are as follows. Every unlisted rank has zero polytopes.*

	rank 3	*rank* 4	*rank* 5	*rank* 6
A_{11}	122	0	0	4
A_{12}	375	173	43	0
A_{13}	3021	202	50	20
A_{14}	8531	255	90	18

In particular the missing ranks four and five for A_{11} are genuine gaps in its rank spectrum.

Proof. For $11 \leq n \leq 14$ take $G = A_n$ in its natural permutation action. We use the classical automorphism theorem $\mathrm{Aut}(A_n) = S_n$ acting by conjugation for $n \geq 7$. Thus the conjugation action of S_n supplies precisely the abstract isomorphisms between markings on G. This standard theorem is a retained input, not a conclusion of the finite search.

The program forms the complete set of nonidentity involutions of A_n from its conjugacy classes. At an accepted prefix it chooses one representative of each orbit of its pointwise stabilizer in S_n on the eligible new involutions, and applies the exact end intersection test. Its cached commuting sets and subgroup tests are the retained common algorithm. The implementation also records prefixes generating other transitive subgroups, and continues extending those prefixes. We retain only the leaves generating A_n for the present claim. These extra records do not merge or duplicate any prefix orbit, and the algorithm stops extending a prefix that already generates A_n.

One implementation detail weakens, without invalidating, the common pruning test. A branch is discarded only when the group generated by the old prefix and every allowed commuting involution is intransitive. Every string extension lies in this group. An intransitive group cannot contain A_n, so no generating A_n marking is lost. The test may retain extra branches, which does not affect completeness or uniqueness. Every appended mark is outside the old generated subgroup. Hence the subgroup-index argument bounds the rank by $\lfloor \log_2 |A_n| \rfloor$, giving the respective complete search limits $24, 27, 31, 35$. The complete runs retain all literal tuples and all zero rows through these limits.

The program records respectively $136, 3935, 3293, 9562$ transitive markings. Exact generation tests select $126, 591, 3293, 8894$ of these as generating $A_{11}, A_{12}, A_{13}, A_{14}$. Grouping those selected tuples by length gives the displayed table. In addition to the recursive enumeration test, a separate execution checks every pair of subsets of the generators of every recorded transitive tuple against the full intersection property. Those executions also check involution orders, string

commutations and transitivity, and repeat the exact generation tests. All pass. These additional checks verify the recorded objects; completeness rests on the retained orbit enumeration proof and the completed full-rank executions.

Ranks one and two cannot occur. A group generated by one involution has order at most two. A group generated by two involutions has a cyclic subgroup of index at most two, generated by their product. Neither can be the nonabelian simple group A_n. This covers the ranks below the enumeration's recording threshold and completes the proof. □

Attribution, finite premises and remaining source scope. The table recovers exactly the original Table 2; it is not a new classification claim. The stabilizer method is the established method described in the source and attributed there to Hartley–Hulpke. Our executable proof makes the extension criterion, pruning and rank coverage explicit and retains the full finite output. All group operations are exact GAP operations and remain software premises. There is no claim of Lean verification.

The source's historical conjecture in Remark 6.1 is separate from these four finite classifications. Its later resolution must not be credited to this computation. Likewise the source's compatible-action theorem, algorithm comparisons, smaller group classifications, sporadic verification and full Co_3 tables are separate obligations. This component alone is not a recovery of the whole source and supplies no whole-original or compression-factor credit.